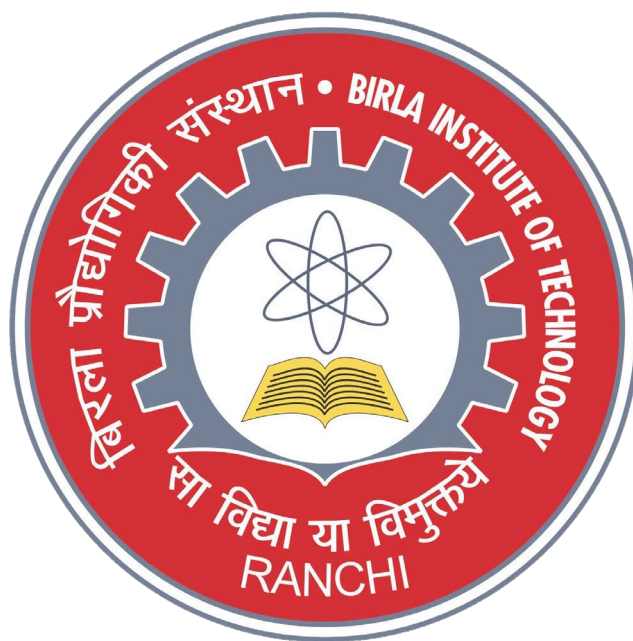


BIRLA INSTITUTE OF TECHNOLOGY



B.Sc. (Computer Science)

CURRICULUM
BASED ON NATIONAL EDUCATION POLICY 2020
(Effective from Academic Session: 2025-26)

DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS

Mr Anurag Joshi

Ms Seema Sharma

Institute Vision

To become a Globally Recognized Academic Institution in consonance with the social, economic and ecological environment, striving continuously for excellence in education, research, and technological service to the National needs.

Institute Mission

- To educate students at Under Graduate, Post Graduate, Doctoral, and Post-Doctoral levels to perform challenging engineering and managerial jobs in industry.
- To provide excellent research and development facilities to take up Ph.D. programmes and research projects.
- To develop effective teaching learning skills and state of art research potential of the faculty.
- To build national capabilities in technology, education, and research in emerging areas.
- To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society.

Department Vision

The department strives to be recognized globally for outstanding education and research, leading to excellent professionals and innovators in the field of Computer Science and Engineering, who can positively contribute to the society.

Department Mission

1. To impart quality education and equip the students with strong foundation that could make them capable of handling challenges of the new century.
2. To maintain state of the art research facilities and facilitate interaction with world's leading universities, industries and research organization for constant improvement in the quality of education and research.

Programme Educational Objectives (PEOs)

PEO 1: The program will produce graduates who will be competent professionals in IT industry, academics, government or entrepreneurs.

PEO 2: Graduates will exhibit professional ethics, critical thinking, problem solving and effective communication skills to work collaboratively in a team-based environment.

PEO 3: The graduates will possess leadership qualities and will be capable of attaining higher positions in their professional career.

PEO 4: Graduates will be able to adapt to the fast-changing world of technology and will become effective professionals to address the technical, social and business challenges.

PEO 5: Graduates will recognize the importance of interdisciplinary learning, engage in lifelong learning and professional development.

(A) Programme Outcomes (POs)

Graduates will be able to:

1. **Discipline knowledge:** Demonstrate the comprehensive knowledge of mathematics, computing fundamentals and domain concepts to enhance their professional skills.



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2. **Problem analysis:** Apply to identify, formulate and analyze solutions to various computing problems using the fundamental principles of computing.
3. **Design/development of solutions:** Ability to design, develop and implement computer-based solutions to real world problems using appropriate tools and techniques.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern IT tools including prediction and modeling to challenging problems.
6. **The graduates and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the IT professionals
7. **Environment and sustainability:** Understand the impact of the professional computer-based solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles to maintain the integrity in a working environment in sustainable societal development through objective, unbiased and truthful actions.
9. **Individual and team work:** Ability to work effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Express thoughts and ideas effectively in understanding computing activities by writing effective reports, making effective presentations, constructing documentation and presenting complex information in a concise manner.
11. **Project management:** Learn to build a project from pre-implementation to completion within constraints in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

(B) Programme Specific Outcomes (PSOs)

1. The ability to analyze, design, code and test application specific or complex problems in Cryptography and Network Security, Design and Analysis of Algorithm, Computer Networks, Cloud Computing, Mobile Computing, Data Mining and Big Data by applying the knowledge of basic sciences, mathematics and fundamentals.
2. The ability to adapt for rapid changes in tools and technology with an understanding of societal and ecological issues relevant to professional practice through life-long learning.
3. Excellent adaptability to function in multi-disciplinary work environment, good interpersonal skills as a leader in a team in appreciation of professional ethics and societal responsibilities.



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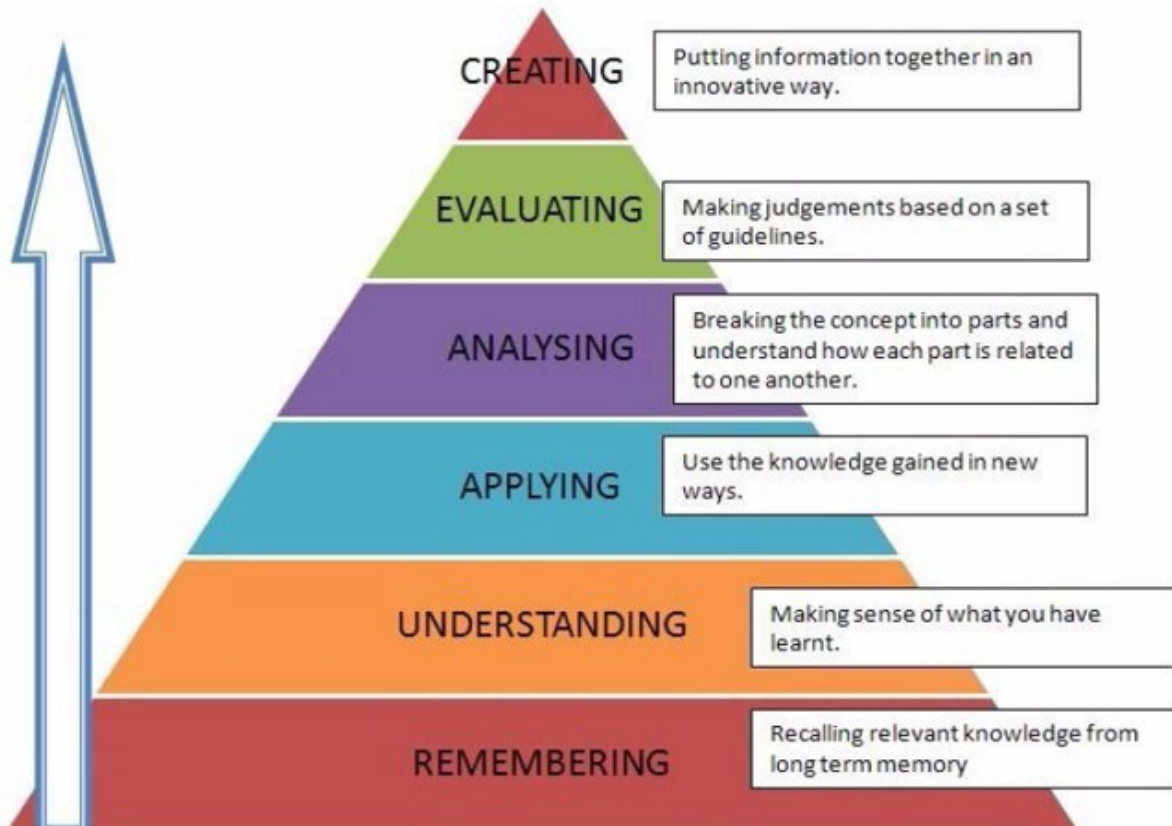


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BLOOM'S TAXONOMY FOR CURRICULUM DESIGN AND ASSESSMENT

Preamble

The design of curriculum and assessment is based on Bloom's Taxonomy. A comprehensive guideline for using Bloom's Taxonomy is given below for reference.



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Program Course Structure

		BIRLA INSTITUTE OF TECHNOLOGY-MESRA, RANCHI COURSE STRUCTURE FOR B.Sc. (Honours / Honours with Research) Computer Science Discipline-Specific Electives in Artificial Intelligence & Machine Learning/Data Science as per NEP-2020 (w.e.f. Academic Session 2025-26) (Proposed)							
Semester/ Session of Study (Recomm ended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T- Tutorial; P-Practical			Total Credits C	
					L	T	P	C	
		THEORY							
First Monsoon	ONE	DSC	SC25103	Structured & Object Oriented Programming	3	0	0	3	
		DSC	SC25105	Operating Systems	3	0	0	3	
		DSC	SC25107	Fundamentals of Computing	2	0	0	2	
		MDC	SC25111	Calculus	3	0	0	3	
		VAC– Elective		VAC Elective I				2	
				VAC Elective II				2	
		SEC-SB Elective		SEC-SB Elective I	2	0	2	3	
		LABORATORIES							
		DSC	SC25104	Structured & Object Oriented Programming Lab	0	0	3	1.5	
		AECC	HS24131	Communication Skills-I	0	0	3	1.5	
TOTAL					21				

DSC: 9.5
VAC: 4
AECC: 1.5
SEC: 3
MDC: 3



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Semester /Session of Study (Recommended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery & credits L-Lecture; T-Tutorial; P-Practical			Total Credits C	
					L (Periods /week)	T (Periods /week)	P (Periods /week)	C	
Second Spring	ONE	THEORY							
		DSC	SC25113	Introduction to Data Structures	3	1	0	4	
		DSC	SC25115	Python Programming	3	0	0	3	
		MDC	SC25117	Mathematics for Computing	3	0	0	3	
		VAC – Elective		VAC Elective III				2	
		LABORATORIES							
		DSC	SC25114	Data Structures Lab	0	0	3	1.5	
		DSC	SC25116	Python Programming Lab	0	0	3	1.5	
		AECC	HS24133	Communication Skills- II	0	0	3	1.5	
		SEC-SB Elective		SEC-SB Elective II	0	1	4	3	
		TOTAL			19.5				
		VOCATIONAL COURSES FOR EXIT AFTER 1 st Year							
Vocational Course I		MN25120	Event Management	3	0	0	3		
Vocational Course II		CA25133	Fundamentals of Multimedia	3	0	0	3		

*Note - Two skill based Vocational courses of 3 credits will be offered at the end of the 1st Year (Level 1) and mandatory for students taking exit after 1st year. However these courses will be optional and non-credit for other students.

DSC: 10

VAC: 2

AECC: 1.5

SEC: 3

MDC: 3

Acronyms Expanded

- AEC : Ability Enhancement Course
- DSC : Discipline Specific Core (Course)
- DSE : Discipline Specific Elective (Course)
- VAC : Value Added Course
- SEC-SB : Skill Enhancement Course-Skill Based
- MDC : Multidisciplinary Course



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Semester/ Session of Study (Recommended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T-Tutorial; P-Practical			Total Credits C
					L (Periods/ week)	T (Period s/week)	P (Periods /week)	C
Third Monsoon	TWO	THEORY						
		DSC	SC25201	Data Analytics and Visualization	3	0	0	3
		DSC	SC25203	Database Design Concepts	3	0	0	3
		MDC	SC25209	Complex Variables and Linear Algebra	3	0	0	3
		MDC	MN25106	Principles of Management	3	0	0	3
		AEC	MN25109	Public speaking and creative writing	1	0	2	2
		SEC-SB		SEC-SB Elective III	2	0	2	3
		LABORATORIES						
		DSC	SC25202	Data Analytics Lab	0	0	3	1.5
		DSC	SC25204	Database Design Lab	0	0	3	1.5
		TOTAL			20			

Semester/ Session of Study (Recomm ended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T-Tutorial; P-Practical			Total Credits
					L (Periods /week)	T (Periods /week)	P (Periods /week)	C
Fourth Spring	TWO	THEORY						
		DSC	SC25213	Data Mining and Applications	3	0	0	3
		DSC	SC25215	Fundamentals of Computer Algorithms	3	1	0	4
		DSC	SC25217	Computer Networks	3	0	0	3
		DSE I (AI&ML) / DSE I (DS)	DSE I		3	1	0	4
		AEC	MN25201	Personality Development	3	0	0	3
		LABORATORIES						
		DSC	SC25214	Data Mining Lab	0	0	3	1.5
		DSC	SC25216	Computer Algorithms lab	0	0	3	1.5
		TOTAL			20			

Total Credits after II year : 40.5.+40=80.5.

Exit option with **Diploma in CS** after completing Vocational Courses
VOCATIONAL COURSES FOR EXIT AFTER 2nd Year

Vocational Course III	MN25214	Basic of Financial Markets and Equity Research	3	0	0	3
Vocational Course IV	CA25225	Business Intelligence	3	0	0	3

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Semester/ Session of Study (Recomm ended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T-Tutorial; P-Practical			Total Credits
					L (Periods /week)	T (Periods /week)	P (Periods /week)	C
Fifth Monsoon	THREE	THEORY						
		DSC	SC25301	Software Engineering	3	0	0	3
		DSC	SC25303	Basics of Machine learning	3	0	0	3
		DSC	SC25307	Cloud Computing	3	0	0	3
		DSE –II (AI&ML)/ DSE- II (DS)	DSE- II		3	0	0	3
		LABORATORIES						
		DSC	SC25302	Software Engineering Lab	0	0	3	1.5
		DSC	SC25304	Machine Learning Lab	0	0	3	1.5
		DSE –II (AI&ML)/ DSE –II (DS)	DSE- II		0	0	3	1.5
		Minor Internship/ Project	SC25308	Minor Project-2 / MOOCS	0	0	0	3
				TOTAL	19.5			
Semester/ Session of Study (Recomm ended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T-Tutorial; P-Practical			Total Credits
					L (Periods /week)	T (Periods /week)	P (Periods /week)	C
Sixth Spring	THREE	THEORY						
		DSC	SC25323	Computer Organization and Architecture	3	0	0	3
		DSC	SC25325	Web Programming	3	0	0	3
		DSC	SC25321	Cyber Security	3	0	0	3
		DSE – III (AI&ML)/ DSE – III (DS)	DSE-III		3	0	0	3
		LABORATORIES						
		DSE –III (AI&ML)/ DSE -III (DS)	DSE-III		0	0	3	1.5
		DSC	SC25326	Web Programming Lab	0	0	3	1.5
		DSC	SC25330	Project	0	0	0	6
			TOTAL	21				

Total Credits after III years :40.5+40+40.5=121

EXIT OPTION WITH B.Sc. (CS)

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Semester/ Session of Study (Recommended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T-Tutorial; P-Practical			Total Credits
					L (Periods /week)	T (Periods /week)	P (Periods /week)	
					THEORY			
Seventh Monsoon	FOUR	DSC	SC25401	Blockchain Technology	3	1	0	4
		DSC	SC25415	Full Stack Application Development	3	0	0	3
		DSC	SC25427	Internet Of Things	3	0	0	3
		DSE – IV (AI & ML)/ DSE- IV (DS)	DSE-IV		3	0	0	3
		DSE-V (AI&M L)/ DSE-V (DS)	DSE-V		3	0	0	3
		LABORATORIES						
		DSC	SC25416	Full Stack Application Development Lab	0	0	3	1.5
		DSC	SC25428	Internet Of Things Lab	0	0	3	1.5
		DSE – IV (AI & ML) DSE – IV (DS)	DSE-IV		0	0	3	1.5
	TOTAL							20.5

Semester/ Session of Study (Recommended)	Course Level	Category of Course	Course Code	Courses	Mode of delivery and credits L-Lecture; T-Tutorial; P-Practical			Total Credits	
					L (Periods /week)	T (Periods /week)	P (Periods /week)	C	
					THEORY				
Eighth Spring	FOUR	DSC	SC25411	Business & Financial Analytics	3	0	0	3	
		DSC		MOOC	3	0	0	3	
		LABORATORIES							
		DSC	SC25412	Business & Financial Analytics Lab	0	0	3	1.5	
		Research Project/ Dissertation	SC25420	Research project /Internship with Viva-voce and seminar presentation.	0	0	0	12	
	TOTAL							19.5	

Total Credits after IV years: 40.5+40+40.5+40=161

- (i) After four years **B.Sc. Honours (CS)** will be given
or

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- (ii) Students who secure 7.5 CGPA and above at the end of sixth semester are eligible to undertake research stream in the fourth year and have to complete Research Project/Dissertation (12 Credits) to become eligible for **B.Sc. Honours with Research (CS) degree**

Acronyms Expanded

- AEC : Ability Enhancement Course
- DSC : Discipline Specific Core (Course)
- DSE : Discipline Specific Elective (Course)
- VAC : Value Added Course
- SEC-SB : Skill Enhancement Course-Skill Based
- MDC : Multidisciplinary Course

Discipline Specific Elective Courses (DSE)

Discipline Specific Elective Courses (DSE I)						
	Course Code		L	T	P	C
	SC25211	Artificial Intelligence and Applications (AI & ML)	3	1	0	4
	SC25221	Data Science Concepts (Data Science)	3	1	0	4
Discipline Specific Elective Courses (DSE II)						
	SC25305	Deep Learning (AI & ML)	3	0	0	3
	SC25311	Exploratory Data Analytics (Data Science)	3	0	0	3
	SC25306	Deep Learning Lab (AI & ML)	0	0	3	1.5
	SC25312	Exploratory Data Analytics Lab (Data Science)	0	0	3	1.5
Discipline Specific Elective Courses (DSE III)						
	SC25315	Generative AI and Applications (AI & ML)	3	0	0	3
	SC25319	Big Data Analytics (Data Science)	3	0	0	3
	SC25316	AI Applications Lab (AI & ML)	0	0	3	1.5
	SC25320	Big Data Analytics Lab (Data Science)	0	0	3	1.5
Discipline Specific Elective Courses (DSE IV)						
	SC25409	Natural Language Processing (AI & ML)	3	0	0	3
	SC25421	Computer Vision (Data Science)	3	0	0	3
	SC25410	NLP Lab (AI & ML)	0	0	3	1.5
	SC25422	Computer Vision Lab (Data Science)	0	0	3	1.5
Discipline Specific Elective Courses (DSE V)						
	SC25405	Research Methodology (AI & ML) AND (Data Science)	3	0	0	3



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Ability Enhancement Course (AEC)

Course Code	Subject	L	T	P	C
HS24131	Communication Skills-1	0	0	3	1.5
HS24133	Communication Skills-2	0	0	3	1.5
MN25109	Public speaking and creative writing	1	0	2	2
MN25201	Personality Development	3	0	0	3
BF25207	Creativity and Critical Thinking Skills	2	1	0	3
BF25213	Effective Writing and Presentation Skills	1	1	0	2
	Total				8

Skill Enhancement Course-Skill Based (SEC-SB)

Course Code	Subject	L	T	P	C
CA25110	Office Automation Tools	0	1	4	3
CA25112	Linux administration	0	1	4	3
CA25130	Programming with MATLAB	0	1	4	3
CA25132	Introduction to Latex	0	1	4	3
CA25209	Statistics with R	2	0	2	3
MN25203	Computerized Accounting	2	0	2	3
SC25121	Quantitative Skills	2	0	2	3
	Total				9

Value Added Courses (VAC)

Course Code	Subject	L	T	P	C
MN25102	Human Values and Professional Ethics	2	0	0	2
HS24211	Indian Knowledge System	2	0	0	2
MN25111	Digital Empowerment	2	0	0	2
SC25169	Design Thinking	2	0	0	2
MN25112	Emotional Intelligence	2	0	0	2
CA25109	Environmental Science	2	0	0	2
MN25103	Yoga	1	0	2	2
	Total				6

Multidisciplinary Course (MDC)

Course Code	Subject	L	T	P	C
SC25111	Calculus	3	0	0	3
SC25117	Mathematics For Computing	3	0	0	3
SC25209	Complex Variables and Linear Algebra	3	0	0	3
MN25106	Principles of Management	3	0	0	3
	Total				9



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